

Microeconomics

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Asymmetric information- Moral
Hazard

Asymmetric information

- **asymmetric Information:** one of the 2 parts of a transaction has more information than the other on the exchanged good or service.
 - Example: a worker on his ability or effort, the owner of an apartment on the quality of the apartment, a manager on the effort he puts in a project
- If information is not equally shared: inefficiency
- 2 types of asymmetric information:
 - **Hidden Characteristics:** Things that one side of a transaction knows about itself that the other side would like to know but does not – e.g. Adverse Selection
 - **Hidden Actions:** Actions taken by one side of an economic relationship that the other side cannot observe– e.g. moral hazard

Moral hazard

- Economic situation:
 - **Principal** delegates a project to an **agent**
 - The action chosen by the agent influences but does not totally determine the outcome
 - Principal and agent have different preferences:
 - The principal prefers a high effort by the agent, while the agent prefers to work with lowest effort
 - Unobservability of actions + difference in preferences give rise to **moral hazard**
 - *Moral hazard: agent takes the less costly action that negatively influences the final outcome.*
 - Inefficiency

Remedies

- What contract may induce the agent to take the principal preferred action?
- Problem: fixed wage- action not observable → agent takes the least costly one
- Remedy: wage based on the observable outcome

Remedies

- Trade-off variable wage
 - Gives incentives to the agent to take high effort(+)
 - Costly for the principal because involves some risk for the agent(-)

example

- Suppose the owner group of GM wants to assume a manager (agent) who will launch the 500X in Dubai

Example

- If the project is a success, they will sell $q = 100$ cars, otherwise $q = 20$
- The success depends also on the effort taken by the manager:
 - A high effort, $e = 1$, leads to a success with probability 0.8
 - A low effort, $e = 0$, leads to a success with probability 0.4
- The principal cannot see the effort taken
- 2 possible contracts:
 - Fixed wage: ω_F
 - Variable wage: if $q=100$, manager gets a high salary, otherwise a low one $\bar{\omega}$ ($\underline{\omega}$)

Example

Assumptions

- The only cost is the wage
- $P_{\text{car}} = 1$
 - ➔ revenues = $p q = q$ = output

Then the risk neutral principal profits are:

$$\pi(e, \omega) = q(e) - \omega$$

Example

- Manager prefers a high salary with a low effort, and he is risk averse:

$$U(e, \omega) = \sqrt{\omega} - \Psi(e)$$

- Manager disutility from high effort is $\Psi = 2$, and $\Psi = 0$ from low effort
- Reservation utility of 2
- What contract will maximize principal profits?

Example

timing:

1. Principal offers a contract (w_h, w_L)
2. manager decides whether to accept
3. If h accepts, he chooses the effort level $e = \{0, 1\}$
4. After the project, output realizes and manager is paid

Example

1-First-Best: solution with observable effort

2-Fixed wage and moral hazard

3-Second-Best: best solution when effort not observable

First-best

What salary would be offered?

- 1) If principal wants to hire the manager who will put a high effort, the salary must satisfy the participation constraint:

$$U = \sqrt{\tilde{\omega}} - 2 \geq 2$$

$$U = \sqrt{\tilde{\omega}} - 2 = 2$$

$$\tilde{\omega} = 16$$

First-Best

Expected profits:

$$E\pi = 0.8 \times 100 + 0.2 \times 20 - 16 = 68$$

➤ 2) If , $e = 0$, principal offers $\tilde{\omega} = 4$

$$E\pi = 48$$

➤ 3) Solution: principal offers $\tilde{\omega} = 16$

Moral hazard

- 1) With a fixed wage, manager puts low effort, then the constraint to be satisfied is:

$$U = \sqrt{\tilde{\omega}} - 0 \geq 2$$

- 2) Then principal offers $\omega = 4$

➤ Expected profits: 48

Second-Best

In order to induce the manager to put high effort, variable wage based on outcomes

➤ What is the least costly variable contract?

Second-Best

- Manager expected utility from high effort

$$EU_{e=1} = 0.8 \times \sqrt{\bar{\omega}} + 0.2 \times \sqrt{\underline{\omega}} - 2$$

- Manager expected utility from low effort

$$EU_{e=0} = 0.4 \times \sqrt{\bar{\omega}} + 0.6 \times \sqrt{\underline{\omega}} - 0$$

- Then in order for the manager to put high effort: **(incentive compatibility constraint)**

$$EU_{e=1} \geq EU_{e=0} \quad \Rightarrow \quad \sqrt{\bar{\omega}} - \sqrt{\underline{\omega}} \geq 5$$

Second-Best

➤ Participation constraint:

$$0.8 \times \sqrt{\bar{\omega}} + 0.2 \times \sqrt{\underline{\omega}} - 2 \geq 2$$

Second-Best

$$0.8 \times \sqrt{\bar{\omega}} + 0.2 \times \sqrt{\underline{\omega}} - 2 = 2$$

$$\sqrt{\bar{\omega}} - \sqrt{\underline{\omega}} = 5$$

Second-Best

Then: $\underline{\omega} = 0$ $\bar{\omega} = 25$

➤ Manager gets:

$$EU = 0.8 \times \sqrt{25} + 0.2 \times \sqrt{0} - 2 = 2$$

➤ Principal:

$$E\pi = 0.8 \times (100 - 25) + 0.2 \times (20 - 0) = 64$$

➤ Expected profits higher than fixed wage!

Second-Best

- Why profits are lower in the second best compared to the first best?
 - Because of risk aversion of the manager
- Observable actions: fixed wage of 16 with certainty with non observable actions he gets 20 in expectations
- $20 - 16 = 4$, is the risk premium(16 is the CE of a lottery with an EV of 20)
 - Premium to get into the lottery